$$\langle \Phi | \mathbf{X} | \Psi \rangle = \int \phi^*(x) x \psi(x) dx.$$

Because $x$ is real, it's easy to see that these two integrals are complex conjugates of each other, and therefore that **X** is Hermitian.

What about the operator **D**? In this case, the two sandwiches are

$$\langle \Psi | \mathbf{D} | \Phi \rangle = \int \psi^*(x) \frac{d\phi(x)}{dx} dx \tag{8.7}$$

and

$$\langle \Phi | \mathbf{D} | \Psi \rangle = \int \phi^*(x) \frac{d\psi(x)}{dx} dx. \tag{8.8}$$

To determine if **D** is Hermitian, we need to compare these two integrals and see if they are complex conjugates of each other. In this form, it's a bit difficult to tell. The trick is to do the second integral by parts. As we explained, integration by parts allows you to switch the derivative from one factor in the integrand to the other, as long as you change the sign at the same time. Therefore, the integral in Eq. 8.8 can be rewritten as

$$\langle \Phi | \mathbf{D} | \Psi \rangle = -\int \psi(x) \frac{d\phi^*(x)}{dx} dx. \tag{8.9}$$

Now, we just need to compare the two expressions in Eqs. 8.7 and 8.9, which turns out to be easy. Because of the minus sign, it's clear that they are definitely *not* complex conjugates of each other. Instead, their relationship is captured by

$$\langle \Psi | \mathbf{D} | \Phi \rangle = -\langle \Phi | \mathbf{D} | \Psi \rangle^*,$$